MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **WEEKLY TEST – 03** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Type Questions — MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If $\cos\theta = \frac{2}{3}$, then $2\sec^2\theta + 2\tan^2\theta - 7$ is equal to: [1]

(A) 0 (B) 1 (C) 3 (D) 4

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 0

Justification: $\sec\theta = \frac{3}{2} \implies \sec^2\theta = \frac{9}{4}$ and $\tan^2\theta = \sec^2\theta - 1 = \frac{5}{4}$. Then $2\left(\frac{9}{4}\right) + 2\left(\frac{5}{4}\right) - 7 = 7 - 7 = 0$.

Q2. If $\cot\theta = \frac{7}{8}$, then the value of $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$ is: [1]

(A) $\frac{49}{64}$ (B) $\frac{64}{49}$

(C) $\frac{7}{8}$ (D) $\frac{8}{7}$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{49}{64}$

Justification: Expression simplifies to $\frac{1-\sin^2\theta}{1-\cos^2\theta} = \frac{\cos^2\theta}{\sin^2\theta} = \cot^2\theta = \left(\frac{7}{8}\right)^2 = \frac{49}{64}$.

Q3. In $\triangle PQR$, $\angle Q = 90^\circ$, $PQ = 12$ cm, and $QR = 5$ cm. The value of $\tan P - \cot R$ is: [1]

(A) 0 (B) $\frac{5}{12}$

(C) $\frac{12}{5}$ (D) $\frac{119}{60}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 0

Justification: $\tan P = \frac{QR}{PQ} = \frac{5}{12}$ and $\cot R = \frac{QR}{PQ} = \frac{5}{12}$. Hence $\tan P - \cot R = 0$.

Q4. Which of the following can **never** be the value of $\sec\theta$ for any acute angle θ? [1]

(A) $\frac{1}{2}$ (B) 2 (C) $\frac{5}{4}$ (D) 3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) $\frac{1}{2}$

Justification: For $0^\circ \le \theta < 90^\circ$, $0 < \cos\theta \le 1$, which implies $\sec\theta = \frac{1}{\cos\theta} \ge 1$. Hence $\frac{1}{2} < 1$ is impossible.

Q5. **Assertion (A):** $\sin(A + B) = \sin A + \sin B$ for all angles A and B.

Reason (R): If $A = 30^\circ$ and $B = 60^\circ$, $\sin(A+B) = \sin 90^\circ = 1$, whereas $\sin 30^\circ + \sin 60^\circ = \frac{1+\sqrt{3}}{2} \neq 1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

Justification: Sine is not a linear function over angle addition; the counterexample in (R) disproves (A). Thus (A) is false and (R) is true.

SECTION B: Short Answer Type-I

Q6. Evaluate: $\dfrac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Substitute Standard Trigonometric Values [1 Mark]

We have $\cos 60^\circ = \frac{1}{2}$, $\sec 30^\circ = \frac{2}{\sqrt{3}}$, $\tan 45^\circ = 1$, $\sin 30^\circ = \frac{1}{2}$, and $\cos 30^\circ = \frac{\sqrt{3}}{2}$.

$$\frac{5\left(\frac{1}{2}\right)^2 + 4\left(\frac{2}{\sqrt{3}}\right)^2 - (1)^2}{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \frac{\frac{5}{4} + \frac{16}{3} - 1}{\frac{1}{4} + \frac{3}{4}}$$

Step 2: Simplify Numerator & Denominator [1 Mark]

$$= \frac{\frac{15+64-12}{12}}{1} = \boxed{\mathbf{\frac{67}{12}}}$$

SECTION C: Short Answer Type-II

Q7. Evaluate: $\dfrac{\sin 30^\circ + \tan 45^\circ - \csc 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Substitute Standard Angle Values [1 Mark]

$$\frac{\frac{1}{2} + 1 - \frac{2}{\sqrt{3}}}{\frac{2}{\sqrt{3}} + \frac{1}{2} + 1} = \frac{\frac{3}{2} - \frac{2}{\sqrt{3}}}{\frac{3}{2} + \frac{2}{\sqrt{3}}} = \frac{3\sqrt{3} - 4}{3\sqrt{3} + 4}$$

Step 2: Rationalize Denominator [2 Marks]

$$= \frac{3\sqrt{3} - 4}{3\sqrt{3} + 4} \times \frac{3\sqrt{3} - 4}{3\sqrt{3} - 4} = \frac{(3\sqrt{3})^2 + 4^2 - 24\sqrt{3}}{(3\sqrt{3})^2 - 4^2} = \frac{27 + 16 - 24\sqrt{3}}{27 - 16} = \boxed{\mathbf{\frac{43 - 24\sqrt{3}}{11}}}$$

Q8. Prove that: $\dfrac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Factor Out $\sin\theta$ and $\cos\theta$ [1 Mark]

$$\text{LHS} = \frac{\sin\theta(1 - 2\sin^2\theta)}{\cos\theta(2\cos^2\theta - 1)}$$

Step 2: Replace 1 by $\sin^2\theta + \cos^2\theta$ [1.5 Marks]

$$\text{LHS} = \tan\theta\left(\frac{\sin^2\theta + \cos^2\theta - 2\sin^2\theta}{2\cos^2\theta - (\sin^2\theta + \cos^2\theta)}\right) = \tan\theta\left(\frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta - \sin^2\theta}\right)$$

Step 3: Cancellation [0.5 Mark]

$$\boxed{\text{LHS} = \tan\theta = \text{RHS} \quad \text{[Hence Proved]}}$$

Q9. Prove that: $(\sin A + \csc A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Expand Squares Using $(x+y)^2 = x^2 + y^2 + 2xy$ [1.5 Marks]

$$\text{LHS} = (\sin^2 A + \csc^2 A + 2\sin A \csc A) + (\cos^2 A + \sec^2 A + 2\cos A \sec A)$$

Step 2: Apply Identities $\sin^2 A + \cos^2 A = 1$, $\csc^2 A = 1 + \cot^2 A$, $\sec^2 A = 1 + \tan^2 A$ [1.5 Marks]

$$\text{LHS} = (\sin^2 A + \cos^2 A) + 2(1) + 2(1) + (1 + \cot^2 A) + (1 + \tan^2 A) = 1 + 2 + 2 + 1 + 1 + \cot^2 A + \tan^2 A$$

$$\boxed{\text{LHS} = \mathbf{7} + \tan^{\mathbf{2}} \mathbf{A} + \cot^{\mathbf{2}} \mathbf{A} = \text{RHS} \quad \text{[Hence Proved]}}$$

SECTION D: Case-Based / Applied Situation — MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. Precision Instrument Flight Path / Runway Glide Slope

During final approach, an aircraft follows a calibrated instrument glide slope. At a particular point in its descent, the aircraft A is at an altitude of 1200 m directly above ground point C. The flight path AB to the touchdown zone B is inclined at an angle of $\theta = 30^\circ$ to the horizontal ground line BC. **Based on the above geometric model, answer the following questions:**

(i) State the trigonometric ratio relating altitude AC and direct flight distance AB. **[1]**

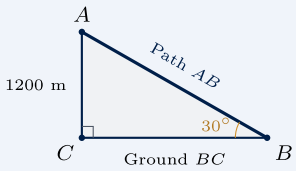
(ii) Calculate the direct straight-line distance AB remaining between the aircraft and the touchdown point. **[1]**

(iii) Calculate the horizontal ground distance BC from the point directly beneath the aircraft to touchdown, and evaluate $\sec^2\theta - \tan^2\theta$. **[2]**

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Sub-part (i): Ratio Connecting AC and AB [1 Mark]

$$\boxed{\sin\theta = \frac{\mathbf{AC}}{\mathbf{AB}}}$$



Sub-part (ii): Direct Flight Distance AB [1 Mark]

$$\sin 30^\circ = \frac{1200}{AB} \implies \frac{1}{2} = \frac{1200}{AB} \implies \boxed{AB = \mathbf{2400}\ \text{m}}$$

Sub-part (iii): Ground Distance BC & Identity Evaluation [2 Marks]

$$\tan 30^\circ = \frac{AC}{BC} \implies \frac{1}{\sqrt{3}} = \frac{1200}{BC}$$

$$\implies \boxed{BC = \mathbf{1200\sqrt{3}}\ \text{m} \approx 2078.46\ \text{m}}$$

$$\sec^2 30^\circ - \tan^2 30^\circ = \left(\frac{2}{\sqrt{3}}\right)^2 - \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{4}{3} - \frac{1}{3} = \boxed{\mathbf{1}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★